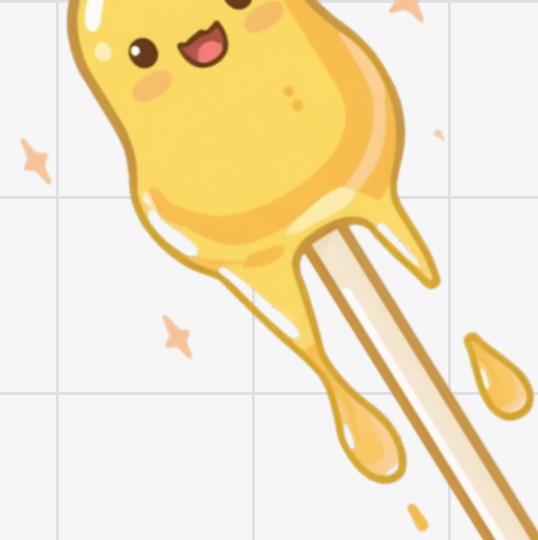
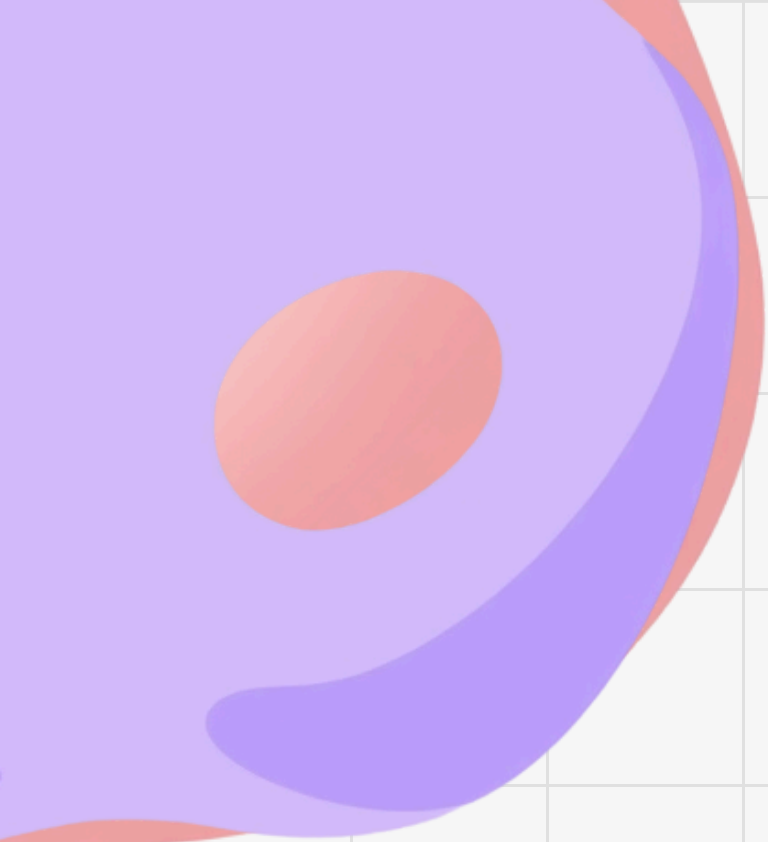
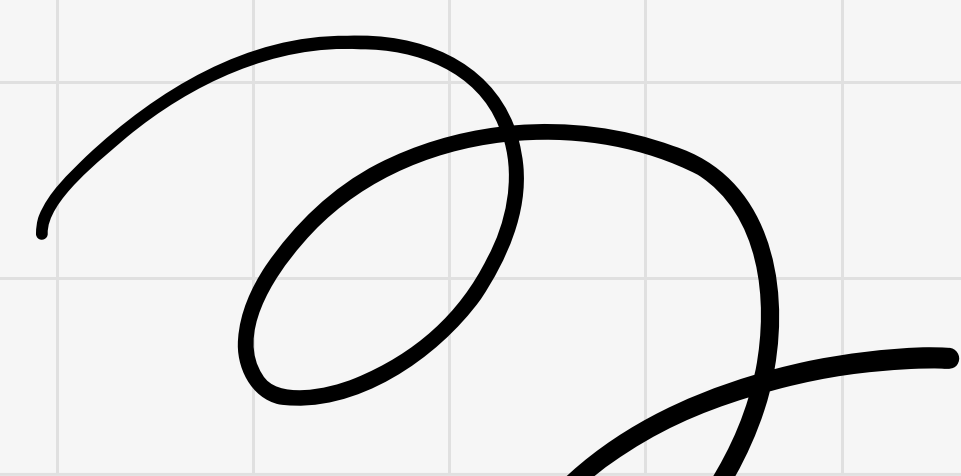
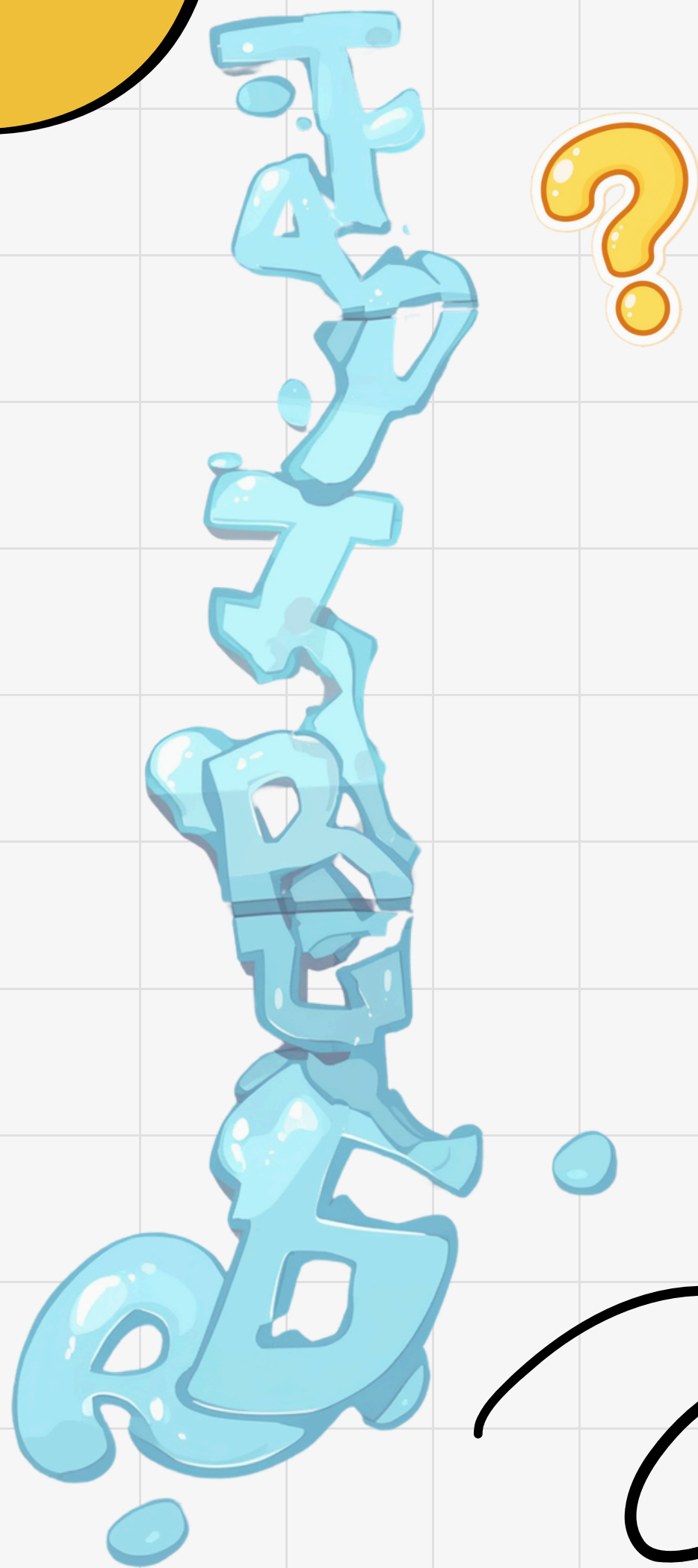




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# Invisible Ink Letter

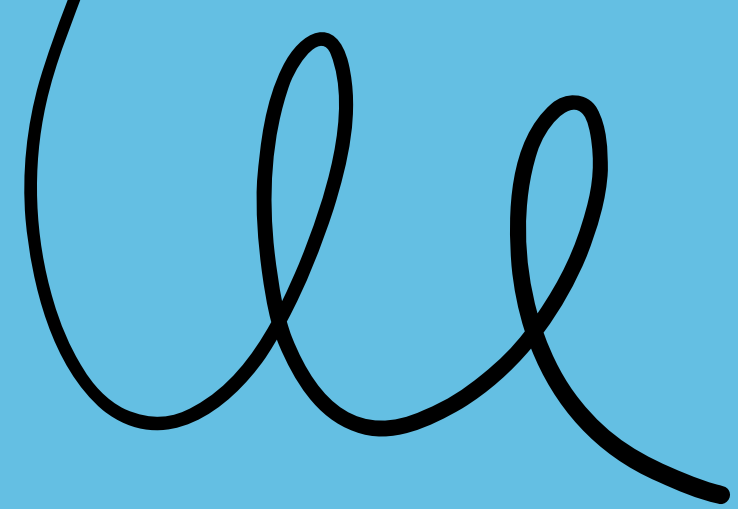




## Warm-Up: Can You See This?

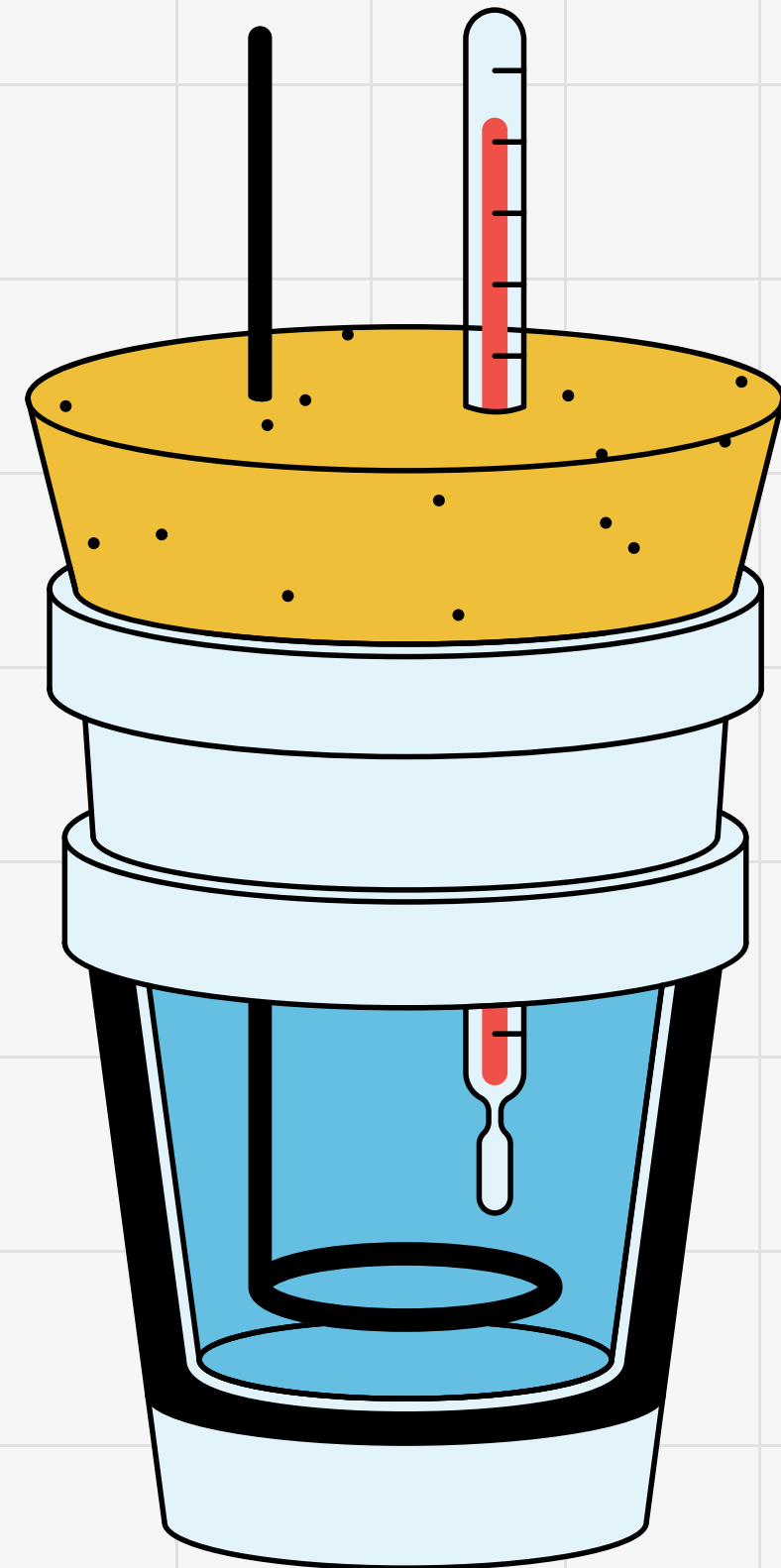
Can you see the water writing  
on the floor?

Sometimes, things are  
**INVISIBLE** — we cannot see  
them!



# Oxidation

When heat or air changes things and they turn brown — like roti browning on a tawa!



# Prediction Time!



What do you think will happen when  
we write with lemon juice and heat it?  
Before experiment, I thought...





# Materials for Our Secret Science Experiment

- Fresh lemons (2-3)
- Cotton buds or thin paintbrush
- White paper for writing
- Teacher uses warm light or hair dryer to reveal
- Secret Scientist badge for you!



# Writing Our Secret Message

1. Dip cotton bud in lemon juice
2. Write a secret message or draw a picture on your paper
3. Let your message dry — it will be INVISIBLE!



## Drying Time - Invisible Ink is Ready!

Now, our secret message is invisible! We cannot see it yet. Can you still see your message? No? That's invisible ink magic!



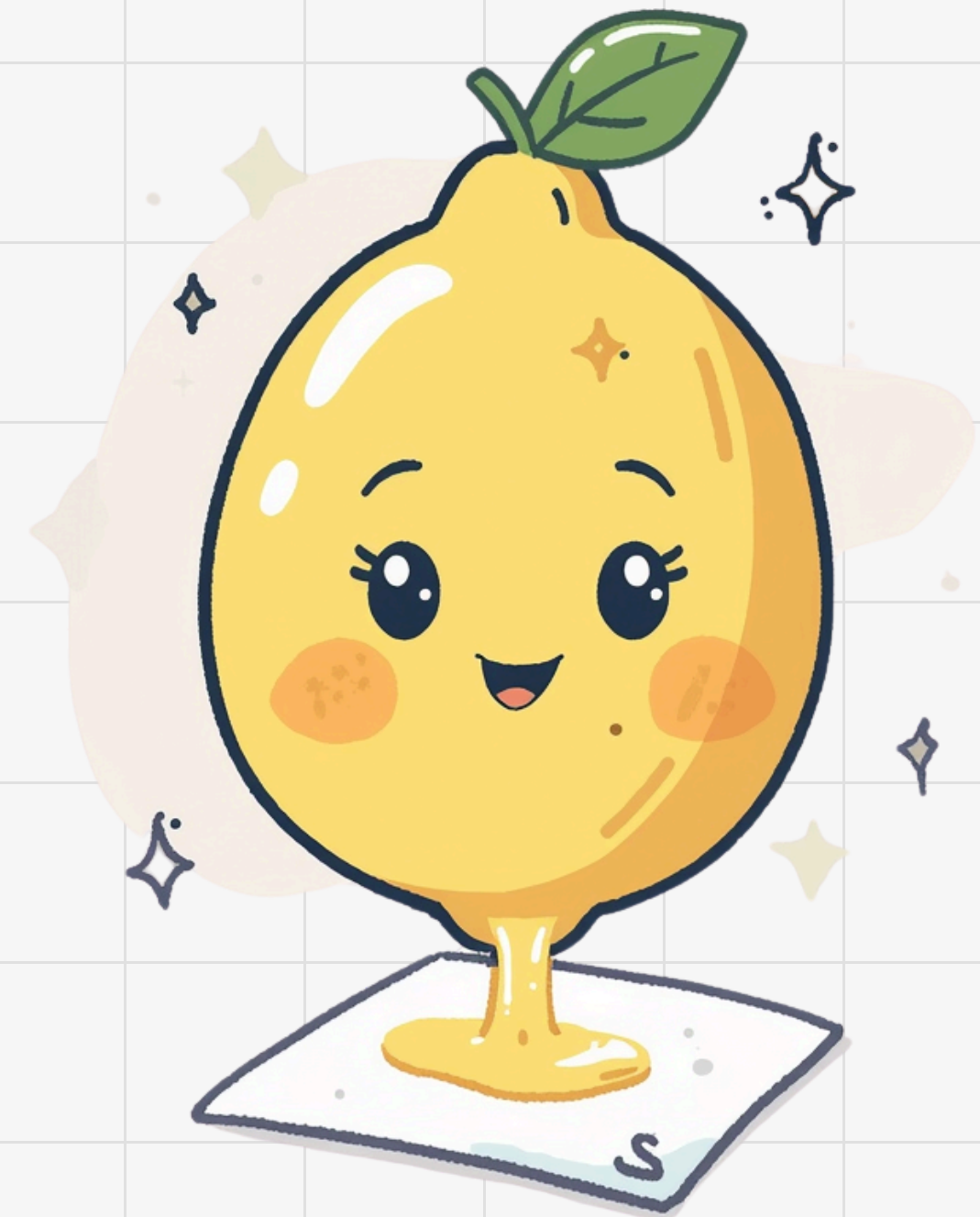
# The Big Reveal!

3... 2... 1... Wow! The secret message appears! Invisible ink turns brown with heat because of OXIDATION!



# What is Invisible Ink?

Lemon juice is a clear liquid called an ACID. When it dries, it is INVISIBLE — we cannot see it. Heat makes the lemon juice turn brown and show the secret! Like writing with water on dry stone floor — it disappears!



# What is Oxidation?

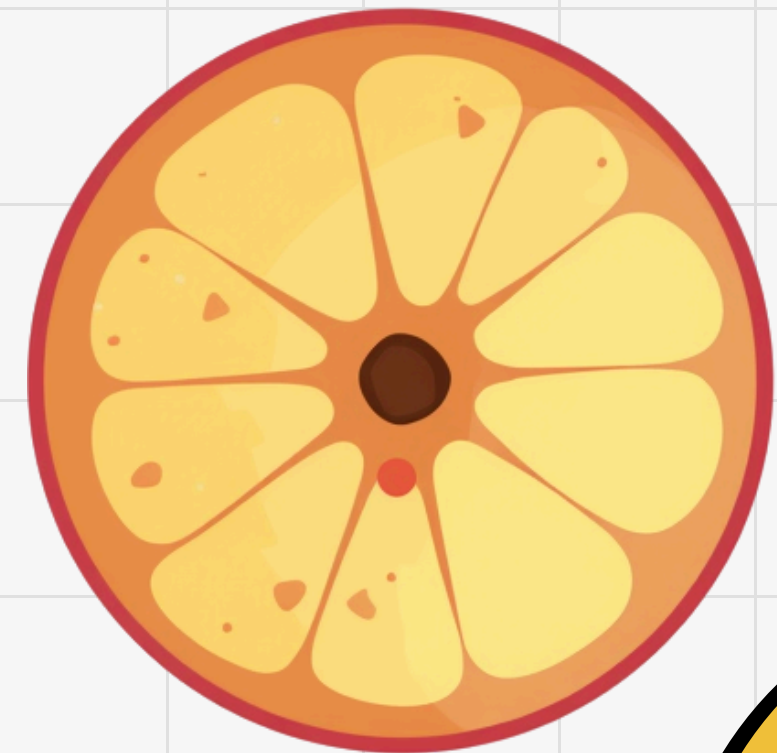
OXIDATION means things change color when they meet heat or air!

- Roti on tawa turns golden-brown
- Cut apple turns brown in air
- Bread toasting becomes golden

Before



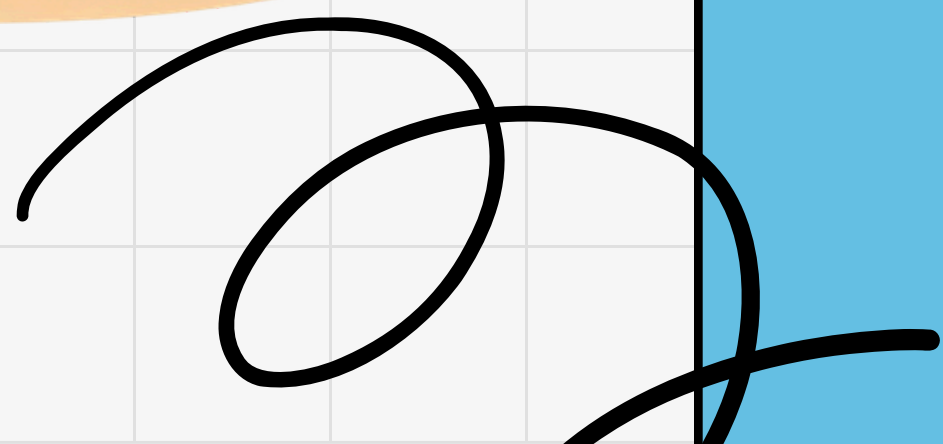
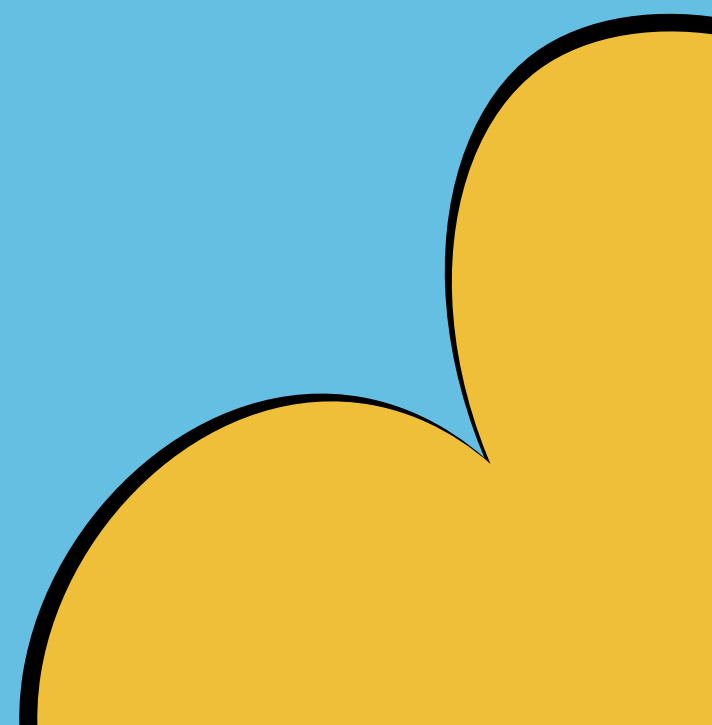
After



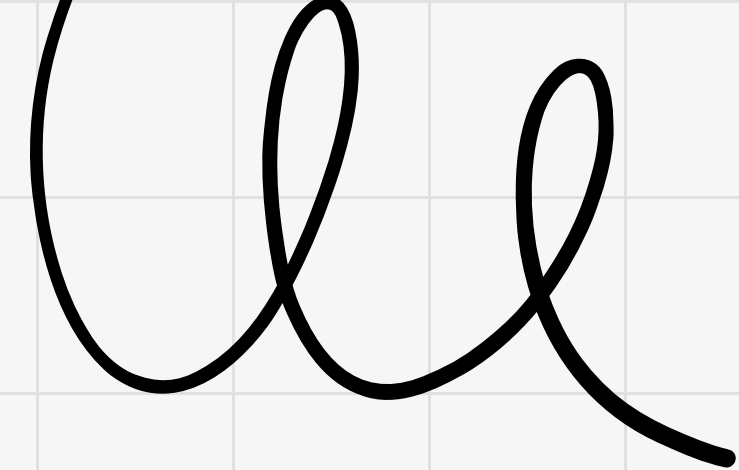
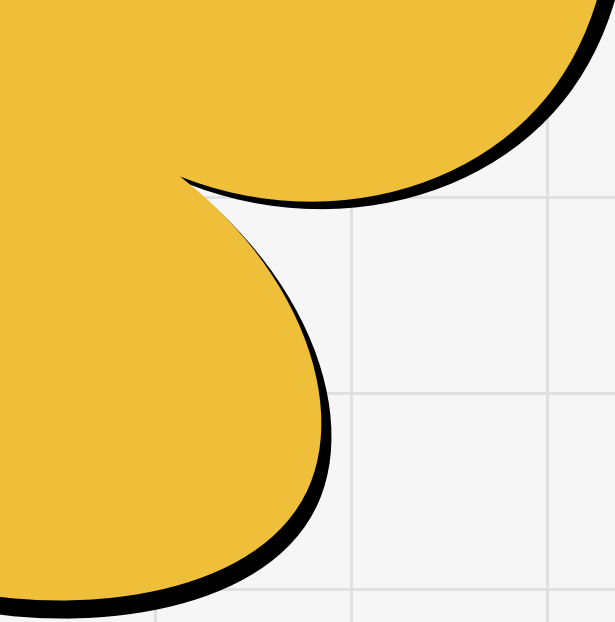


# What is Acid?

Acids are sour liquids with special powers! Lemon juice is an acid called citric acid. Acids are like secret agents hiding special powers!  
Indian examples: Nimbu in dal or salad, Imli in sambar!



# Heat - Energy That Changes Things

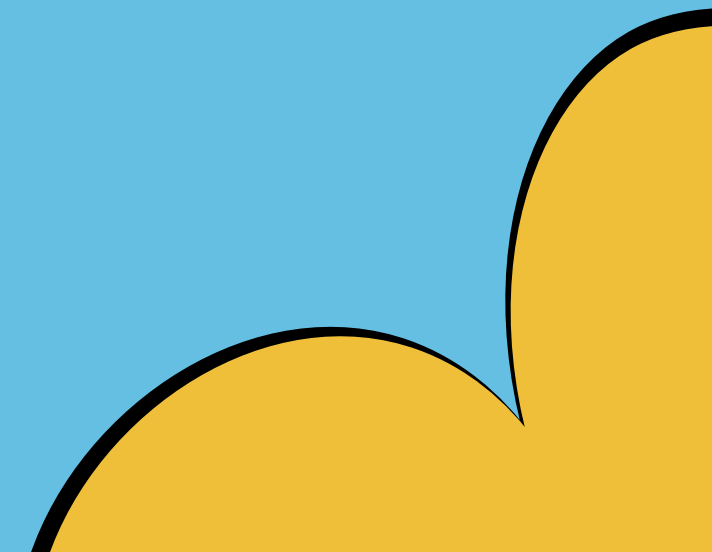


Heat makes things change color, cook, and dry!

☀️ Wet clothes dry in the sun

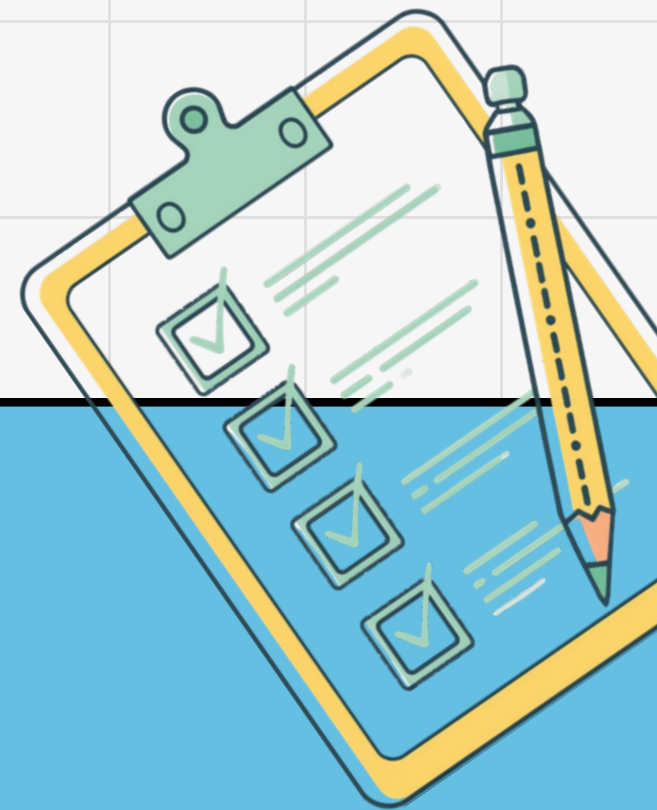
🪔 Diya flame warms the air on Diwali

⚠️ Safety: Only the teacher uses warm light or hair dryer. No open flames!





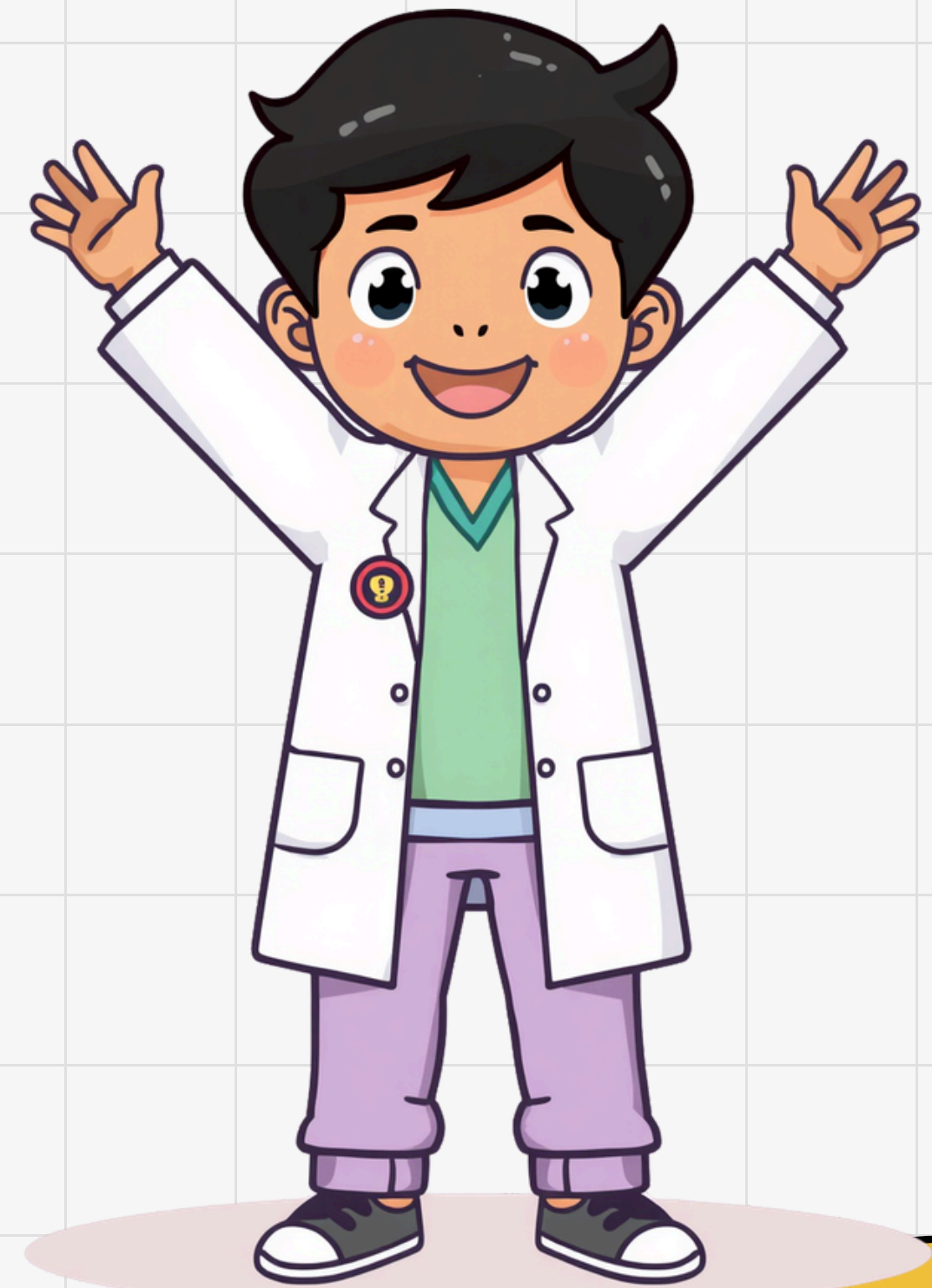
# Record Your Secret Science!



Draw your secret message, write your  
prediction and observation, and  
connect to real life!

# Wrap-Up & Reflection

We learned about INVISIBLE ink that shows with heat!  
We saw how ACIDS like lemon juice have special powers!  
We discovered OXIDATION changes color with heat and air!  
We connected science to roti, apples, and more!  
You are all Secret Scientists now!



# Success & Secret Scientist Badge!



You did it! Everyone made a prediction, used the word INVISIBLE in a sentence, and connected to real-life examples like roti and apple. You used the word ACID too! Earn your Secret Scientist badge!





# What's Next in Our Science Adventure?

Get ready for more fun  
science experiments  
coming soon!

## Fun Science Experiments



1 Orange Juice .. scice

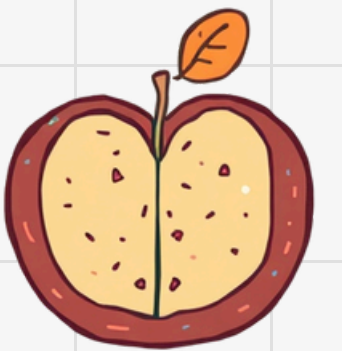
2 Brown Slices .. apless

3 Teate Blices .. science

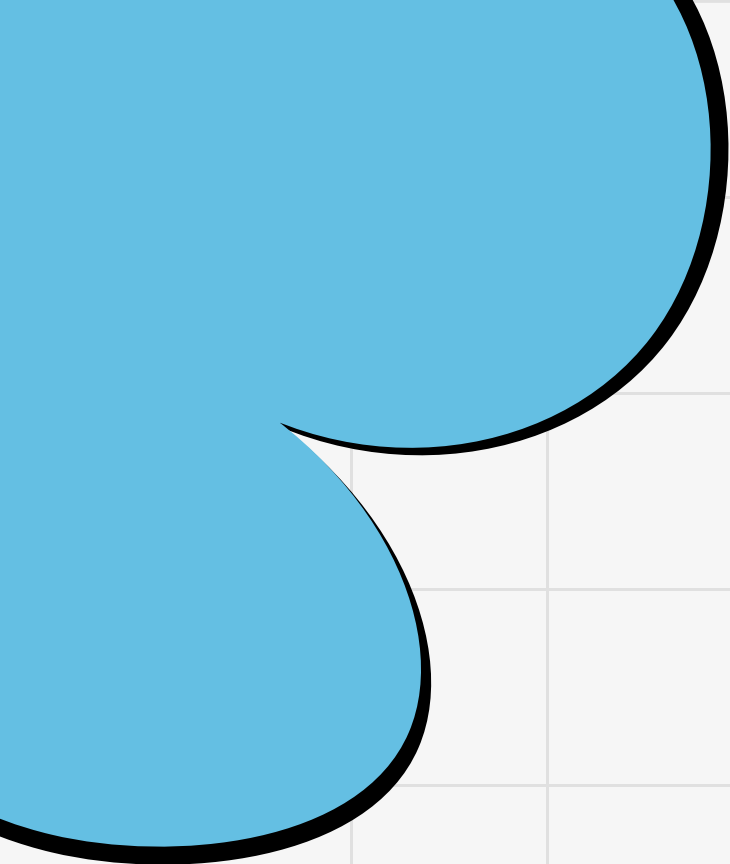
4 Gond Rractey .. chanes

3 Sour fjsnd Bittter

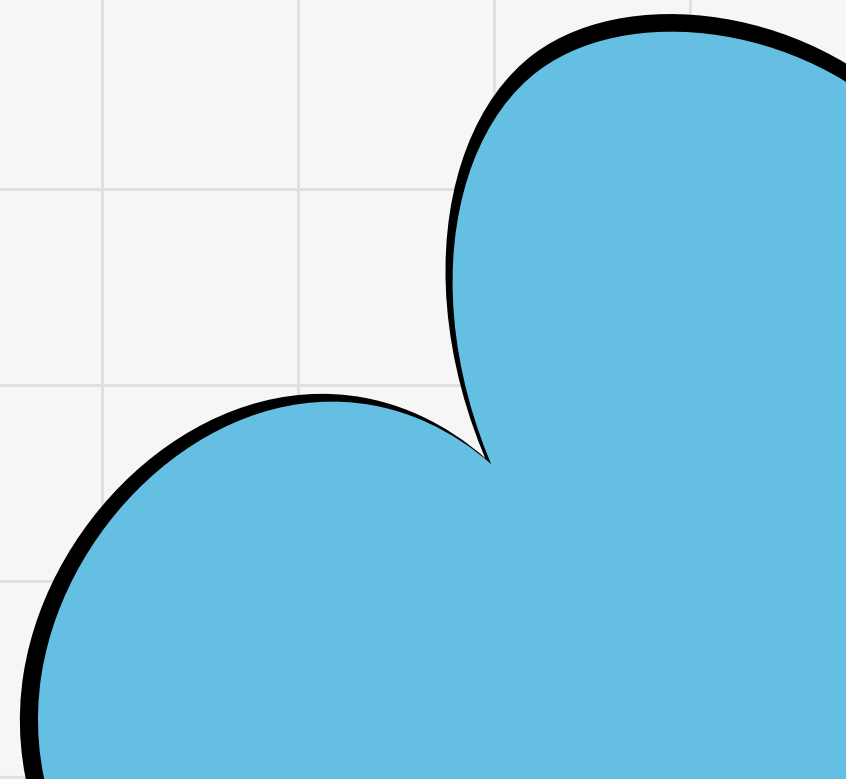
4 Volcano our .. andble







# Thank You!



Thank you for exploring invisible ink with us! Any questions?